

Ollama: The Gateway to Local GenAI

Run powerful open-source LLMs on your own hardware.
No API costs. Total privacy. Zero friction.

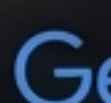


The Paradigm Shift: Renting vs. Owning

PROPRIETARY (The Walled Garden) 🔒



 **OpenAI**
ChatGPT

 **Gemini**
Gemini

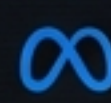
- > Rented Intelligence
- > Pay-per-token API costs
- > Data leaves your premises
- > Black Box Architecture



The Finished Dress
(Fixed Utility).

OPEN SOURCE (The Wild West) 🔓



 **Meta**
Llama



Mistral AI



DeepSeek



Qwen

- \$ Owned Infrastructure
- \$ Free Model Weights
- \$ Local Execution (Air-gapped)
- \$ Fully Customisable



The Raw Cloth
(Infinite Potential).

The Friction & The Solution

The Old Way (The Monster)



Manual weight storage.
Driver incompatibility.
Complex quantization.

The Ollama Way (The Hero)



The 'WhatsApp' for LLMs.
Handles backend complexity
automatically.

The Result

Hello, how can
I help? _

- ✓ **Privacy:** Data stays local.
- ✓ **Latency:** Instant response, no network lag.
- ✓ **Offline:** Works without WiFi.
- ✓ **Cost:** Zero monthly bills.

One-Click Infrastructure: The CLI Experience

If you can type one line, you can run an LLM.

```
ollama — 80x24
[user@macbook ~ % ollama run llama3.2
pulling manifest
pulling 8eeb52dfb3bb... 100% |████████████████████████████████████████| 2.0 GB
verifying sha256 digest
writing manifest
success
>>> Send a message (/? for help)
>>> Why is the sky blue?
The sky is blue due to Rayleigh scattering. Sunlight
reaches Earth's atmosphere and is scattered in all
directions...
```

Magic Commands

ollama pull [model]

Download weights without running.

ollama list

View local inventory.

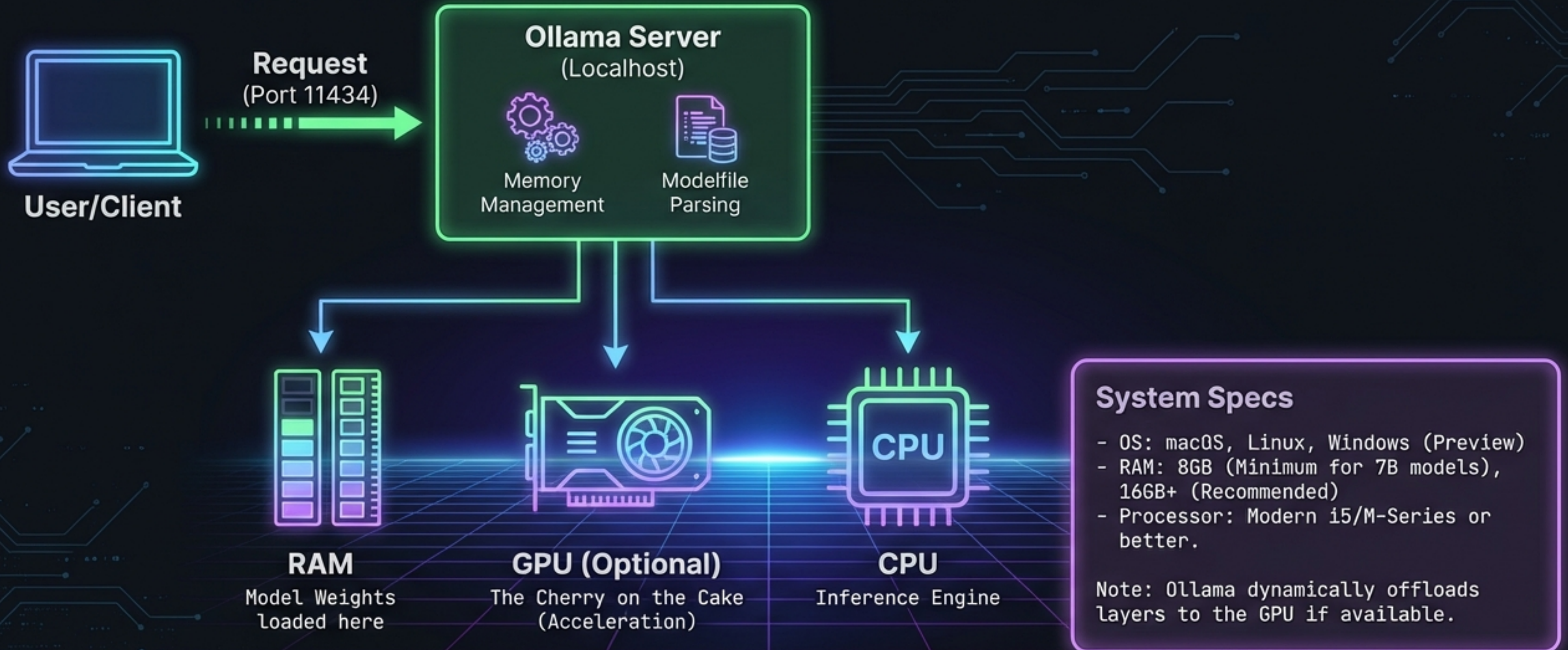
ollama rm [model]

Delete model to free space.

ollama run [model]

Download + Chat instantly.

Architecture & Hardware Requirements



The Library: A Model for Every Use Case

Just like an App Store, but for Intelligence.



Llama 3.2

(JetBrains Mono)

General Purpose

Balanced, Fast,
Reliable.



Mistral / Mixtral

(JetBrains Mono)

Efficiency

High performance,
smaller footprint.



DeepSeek R1

(JetBrains Mono)

Reasoning

Thinking models,
displays Chain of
Thought.

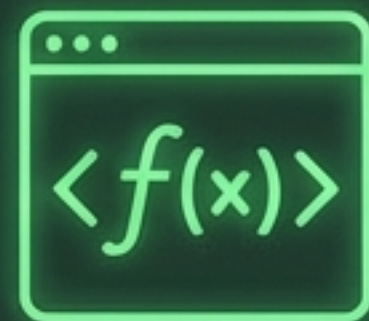


LLaVA

(JetBrains Mono)

Vision / Multimodal

Image recognition
and analysis.



Qwen

(JetBrains Mono)

Coding / Tool Use

Optimized for
mathematics and
function calling.

Parameter Sizes: Models come in 1B, 7B, 70B sizes. Choose based on your available RAM.

Moving to Code: The Python Library

Transition from CLI experimentation to production apps.

Standard
Python Library.

```
import ollama
```

```
# The Chat Method (Maintains History)
```

```
response = ollama.chat(
```

```
    model='llama3.2',
```

```
    messages=[
```

```
        {
```

```
            'role': 'user',
```

```
            'content': 'Why is the sky blue?'
```

```
        }
```

```
    ]
```

```
)
```

```
# Extracting the content
```

```
print(response['message']['content'])
```

Context-aware method
(unlike simple generation).

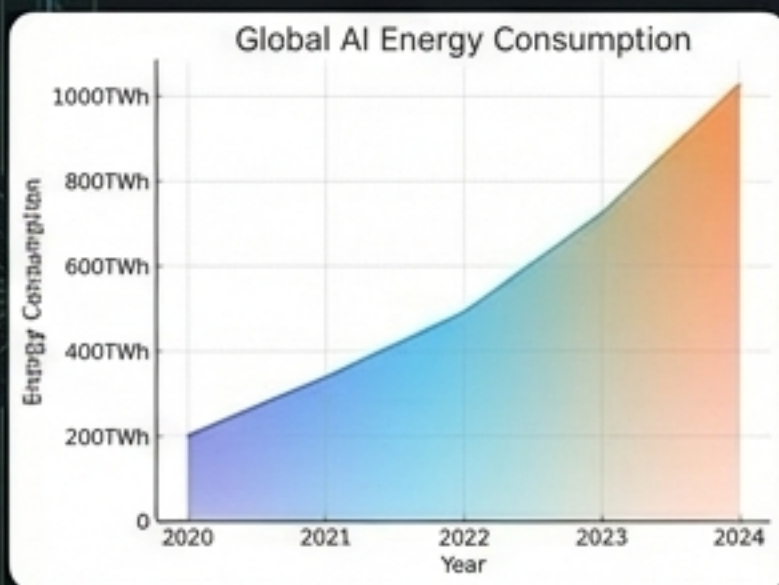
Specify any model
from your library.

Key Methods:

- generate(): Single prompt completion.
- chat(): Structured conversation object.
- stream=True: Enable typewriter effect (token-by-token).

Multimodal Power: Vision Capabilities

Input



Input Image
(JPEG/PNG)

Process

```
with open('chart.jpg', 'rb') as file:  
    response = ollama.chat(  
        model='llava',  
        messages=[  
            {'role': 'user',  
             'content': 'Summarise this chart.',  
             'images': [file.read()]}  
        ]  
    )
```

Output

"The chart shows a steep upward trend in energy consumption by data centres, rising from 200TWh in 2020 to..."

Technical Requirement: Images are passed as Base64 encoded bytes.

Customisation: The Modelfile

Create specialised agents without retraining.

```
FROM llama3.2
```

```
# Set the Persona
```

```
SYSTEM """
```

```
You are a Sentiment Analysis API.  
You only output JSON.  
Do not chat or explain.
```

```
"""
```

```
# Tune Creativity
```

```
PARAMETER temperature 0.1
```

ollama create json-bot
-f Modelfile

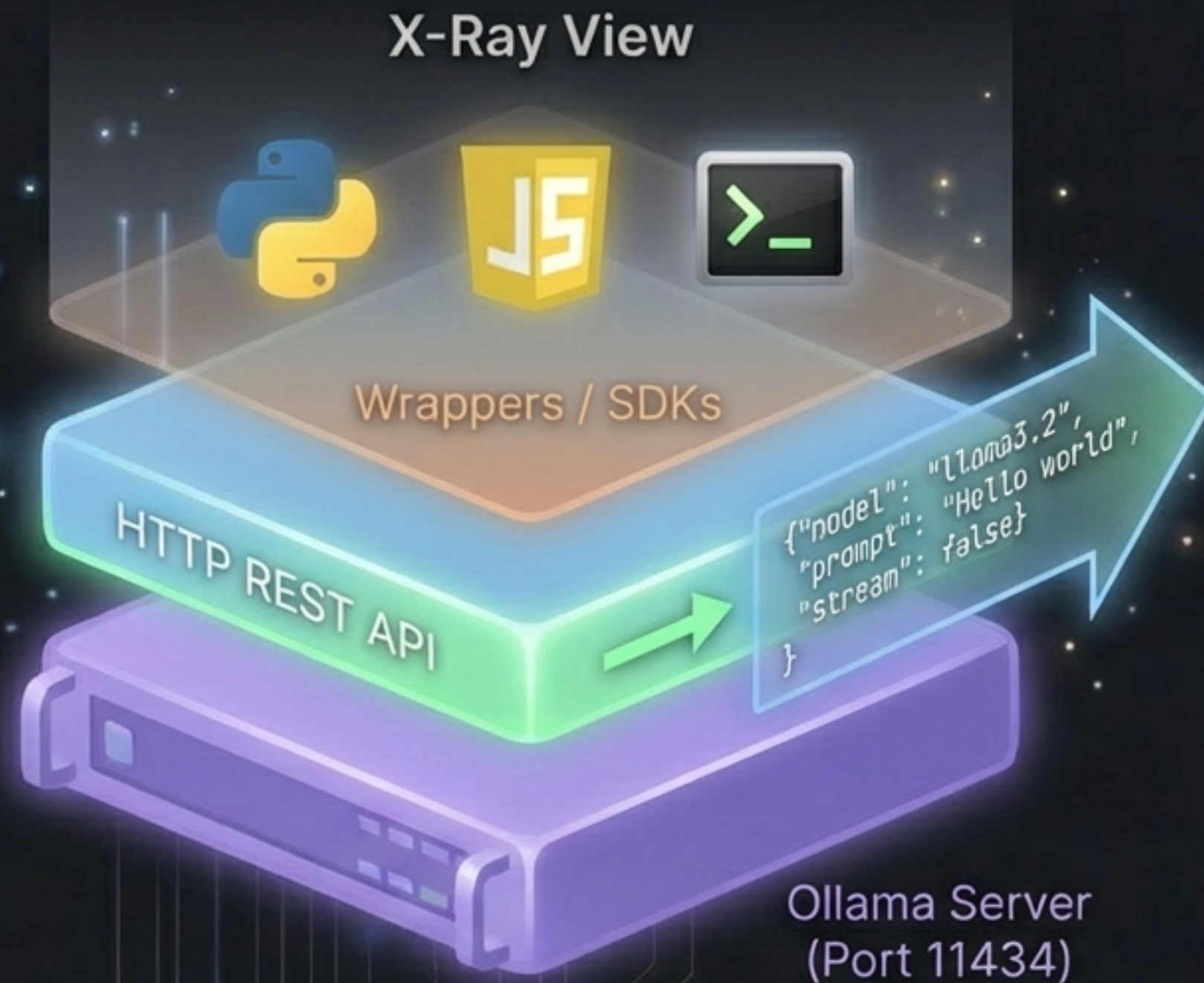


```
>>> I love this course!  
{"sentiment": "positive", "score": 0.98}
```

Key Insight Box: The Cloth Metaphor Applied: We took the generic 'fabric' (Llama 3.2) and stitched it into a specific 'uniform' (JSON Analyzer).

Under the Hood: The REST API

Universal compatibility with any programming language.



bash

[STATUS: ACTIVE]

```
curl http://localhost:11434/api/generate -d '{
  "model": "llama3.2",
  "prompt": "Hello world",
  "stream": false
}'
```

- **Language Agnostic:** Use Node.js, Rust, Go, or Ruby.
- **Stateless:** Perfect for microservices.
- **Direct Control:** Inspect raw JSON payloads.

[STATUS: ACTIVE]

100% NotebookLM

Agentic Workflows: Tool Calling

Giving the LLM 'Hands' to perform actions.

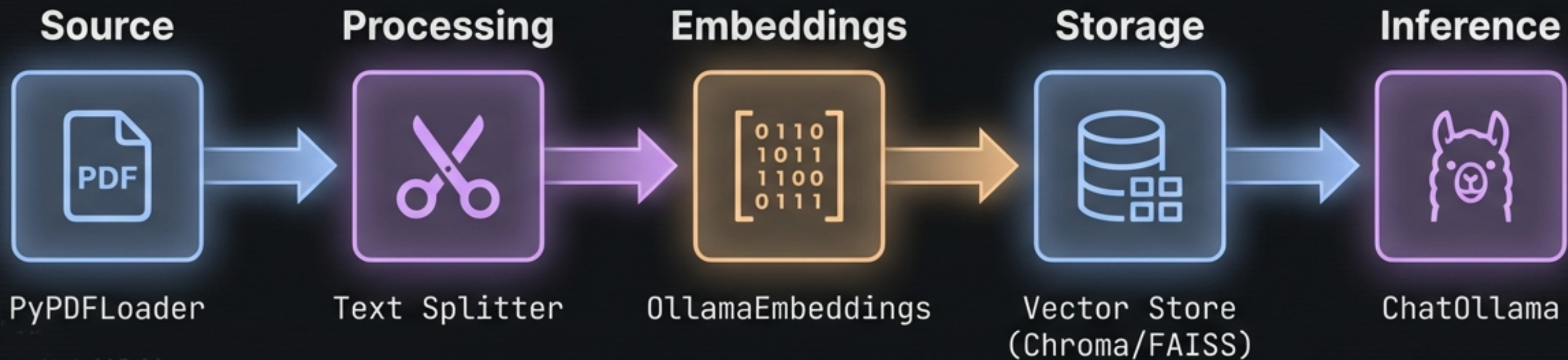


Prerequisites:

1. Tool-capable model (e.g., Qwen, Llama 3.1).
2. Defined JSON Schema for the function.

Orchestration: Integrating with LangChain

Building RAG (Retrieval-Augmented Generation) Pipelines.



LangChain acts as the General Contractor, managing the different workers (Ollama models) to build a complex 'Chat with PDF' application.

Scaling Up: Ollama Cloud

When local hardware hits the ceiling.



Local Machine

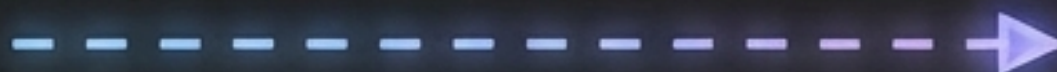
Status:
16GB RAM

Model:
Llama 3.2 (8B)
JetBrains Mono

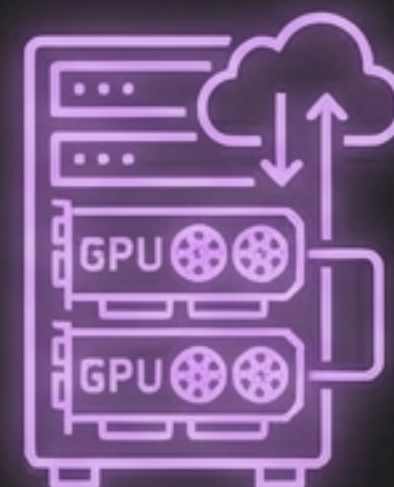
Privacy & Air-gapped.

ollama signin

Authenticate.



Tether: Offload compute to cloud



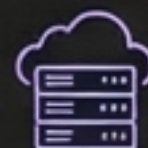
Ollama Cloud

Status:
Data Centre GPU

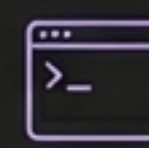
Model:
DeepSeek (671B)
JetBrains Mono

Maximum Intelligence.

Offload compute
to cloud.



Receive response
via same CLI/API.



Note: Data leaves your premises for processing (Privacy Trade-off).

Your Ecosystem Cheat Sheet

Experiment & Chat



```
> ollama run llama3
```



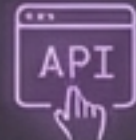
CLI / Ollama App
JetBrains Mono

User: Non-coders, Quick testing.

Build Applications



```
import ollama  
response = ollama.chat(...)
```



Python Library / REST API
JetBrains Mono

User: Developers, Integrators.

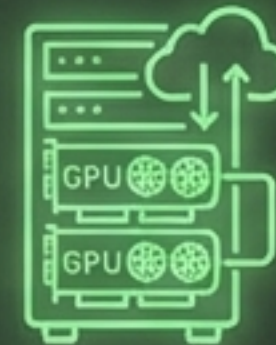
Complex Workflows (RAG)



LangChain
JetBrains Mono

User: AI Engineers, Enterprise Apps.

Massive Scale



600B+ Parameter Models

Ollama Cloud
JetBrains Mono

User: Needs 600B+ Parameter models.

Continue the Journey

This was just the trailer. Now build the movie.



Course: **Generative AI using Open Source LLMs**

Instructor: **Ajay**

Duration: **18-20 Hours**

Content: Deep dives into deployment, fine-tuning, and end-to-end projects.

[Enrol Now](#) | Use link for ₹200 Discount

Liberate your AI.